import pandas as pd import

numpy as np import

matplotlib.pyplot as plt import

seaborn as sns

Load the data

df = pd.read_csv(r'C:\Users\hp\OneDrive - vitbhopal.ac.in\Desktop\DataScienceCodes\banking_data.csv')

Show the first few rows of the data to understand its structure print(df.head())

Age distribution

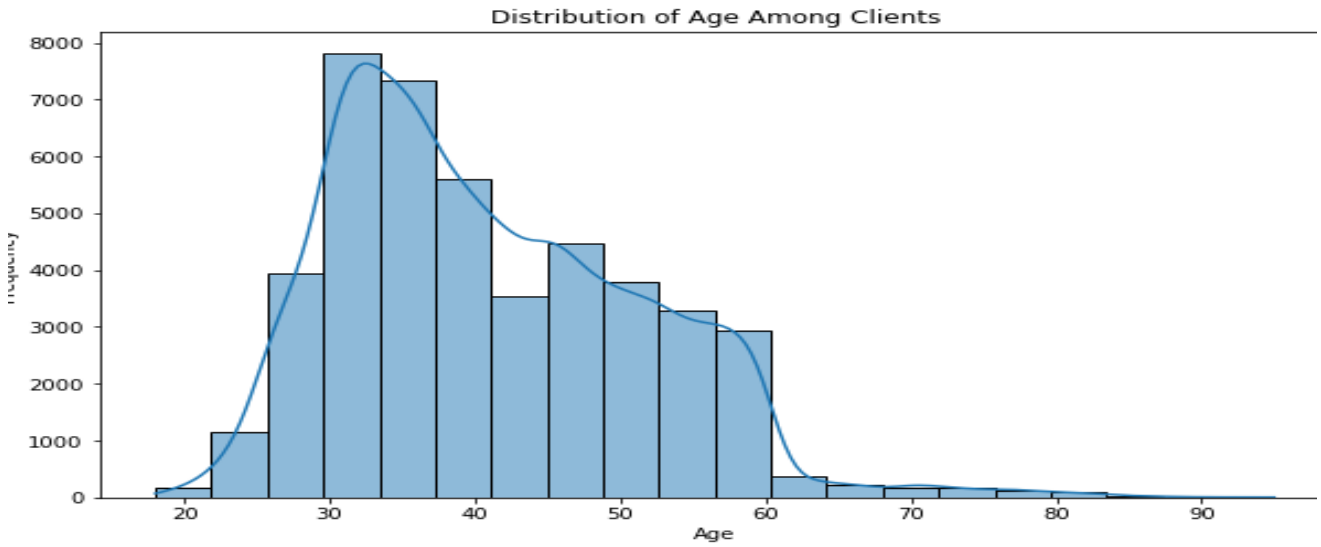
plt.figure(figsize=(10, 6))

sns.histplot(df['age'], bins=20, kde=True)

plt.title('Distribution of Age Among Clients')

plt.xlabel('Age') plt.ylabel('Frequency')

plt.show()

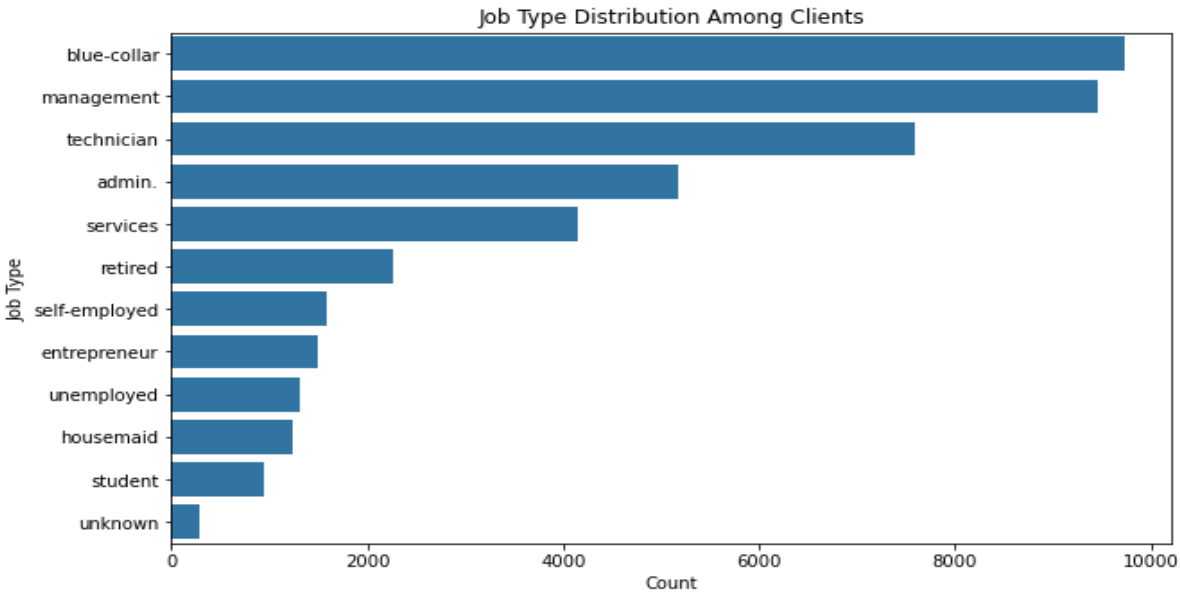


Job type distribution

plt.figure(figsize=(10, 6)) sns.countplot(y='job', data=df,

order=df['job'].value_counts().index) plt.title('Job Type Distribution

Among Clients') plt.xlabel('Count') plt.ylabel('Job Type') plt.show()



Marital status distribution

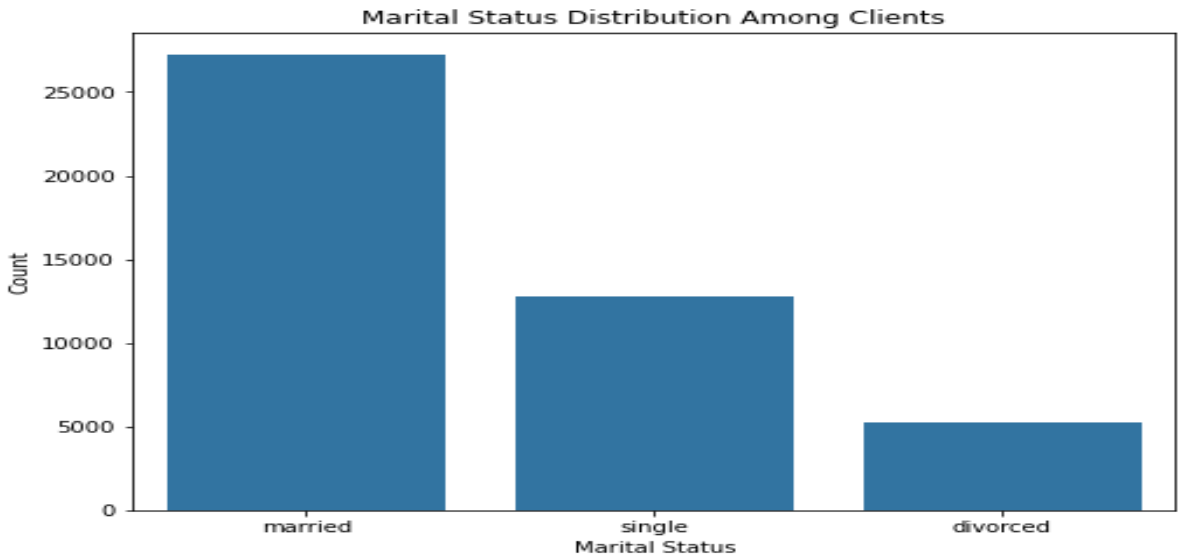
plt.figure(figsize=(8, 6))

sns.countplot(x='marital', data=df)

plt.title('Marital Status Distribution Among

Clients') plt.xlabel('Marital Status')

plt.ylabel('Count') plt.show()

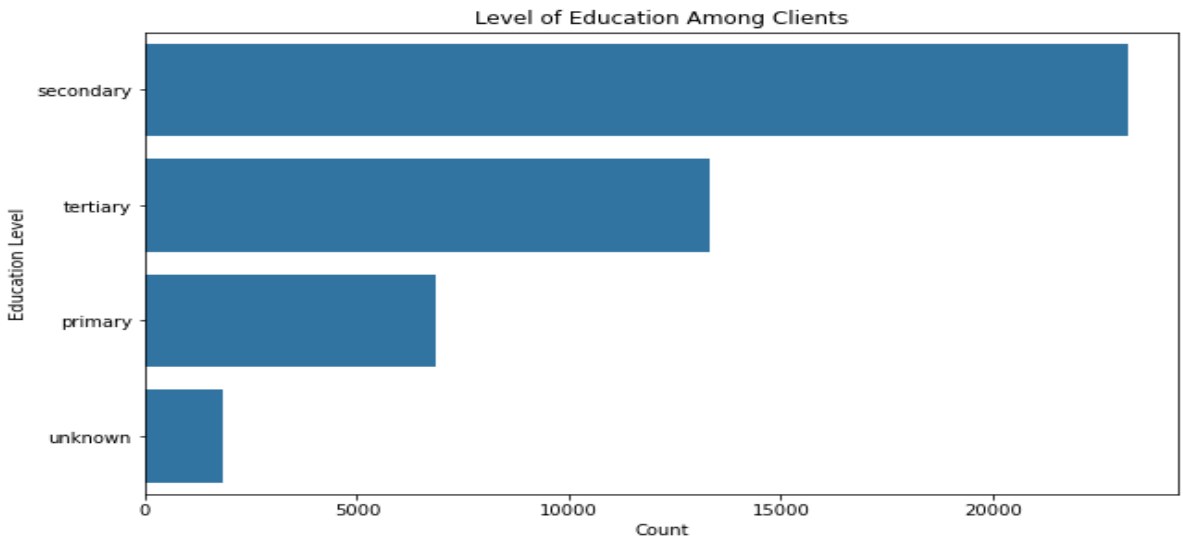


Education level distribution

plt.figure(figsize=(10, 6)) sns.countplot(y='education', data=df,

order=df['education'].value_counts().index) plt.title('Level of Education Among

Clients') plt.xlabel('Count') plt.ylabel('Education Level') plt.show()



Proportion of clients with credit in default

credit_default_proportion = df['default'].value_counts(normalize=True) * 100

print('Proportion of clients with credit in default:\n', credit_default_proportion)

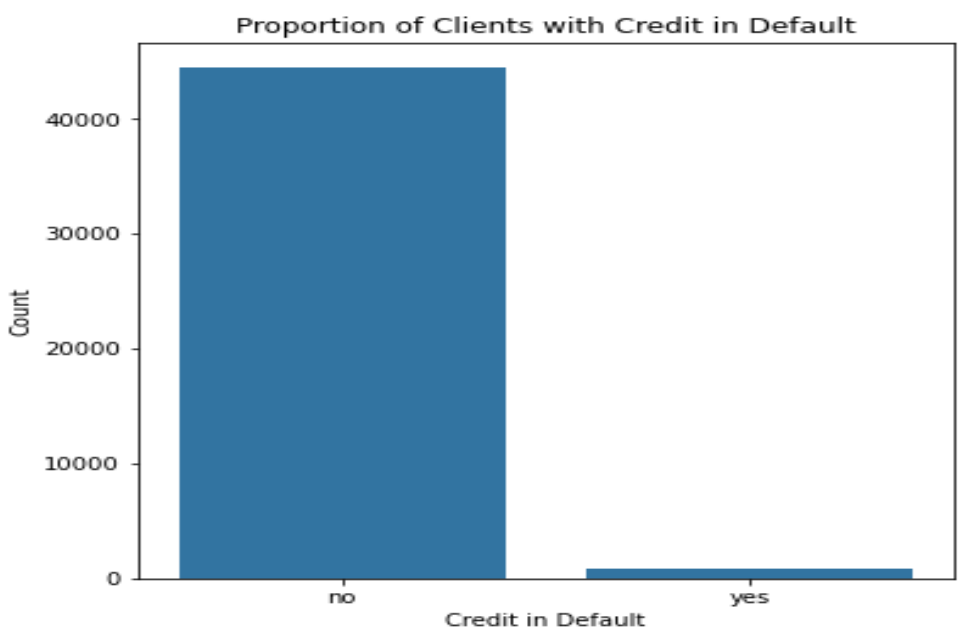
Plotting plt.figure(figsize=(6, 6))

sns.countplot(x='default', data=df)

plt.title('Proportion of Clients with Credit in Default')

plt.xlabel('Credit in Default') plt.ylabel('Count')

plt.show()



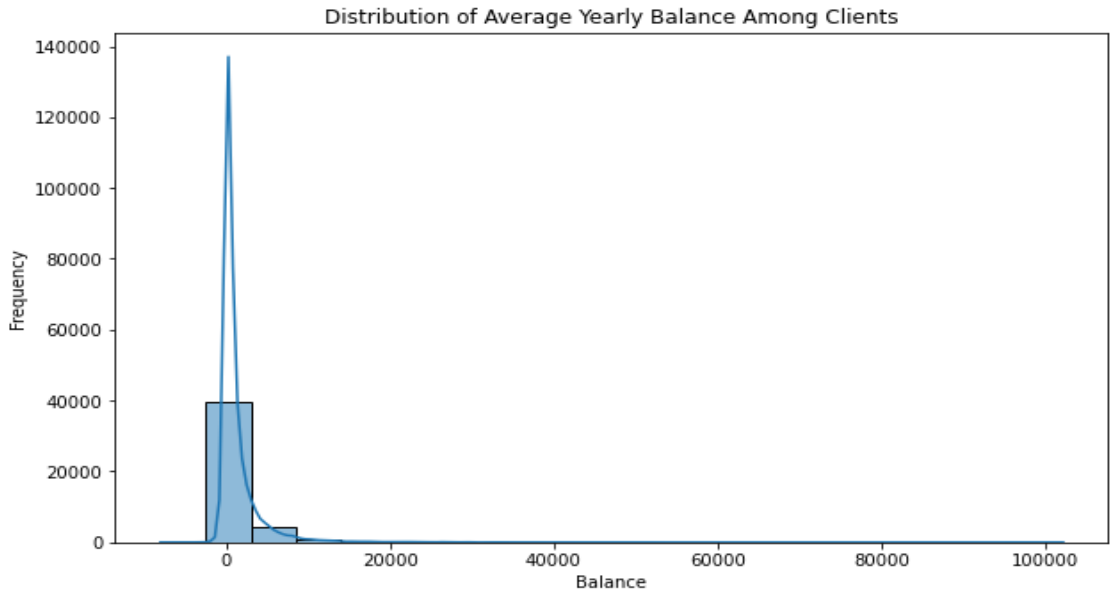
Balance distribution

plt.figure(figsize=(10, 6)) sns.histplot(df['balance'], bins=20,

kde=True) plt.title('Distribution of Average Yearly Balance

Among Clients')plt.xlabel('Balance') plt.ylabel('Frequency')

plt.show()



Housing loan distribution

housing_loan_counts = df['housing'].value_counts() print('Number of clients with housing loans:\n', housing_loan_counts)

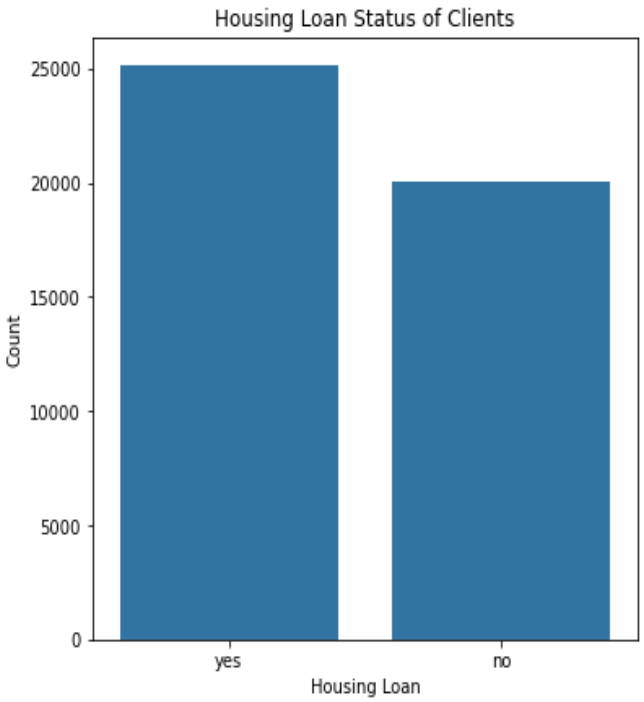
Plotting plt.figure(figsize=(6, 6))

sns.countplot(x='housing', data=df)

plt.title('Housing Loan Status of Clients')

plt.xlabel('Housing Loan')

plt.ylabel('Count') plt.show()



Personal loan distribution

personal_loan_counts = df['loan'].value_counts() print('Number of

clients with personal loans:\n', personal_loan_counts)

Plotting plt.figure(figsize=(6, 6))

sns.countplot(x='loan', data=df)

plt.title('Personal Loan Status of Clients')

plt.xlabel('Personal Loan')

plt.ylabel('Count') plt.show()

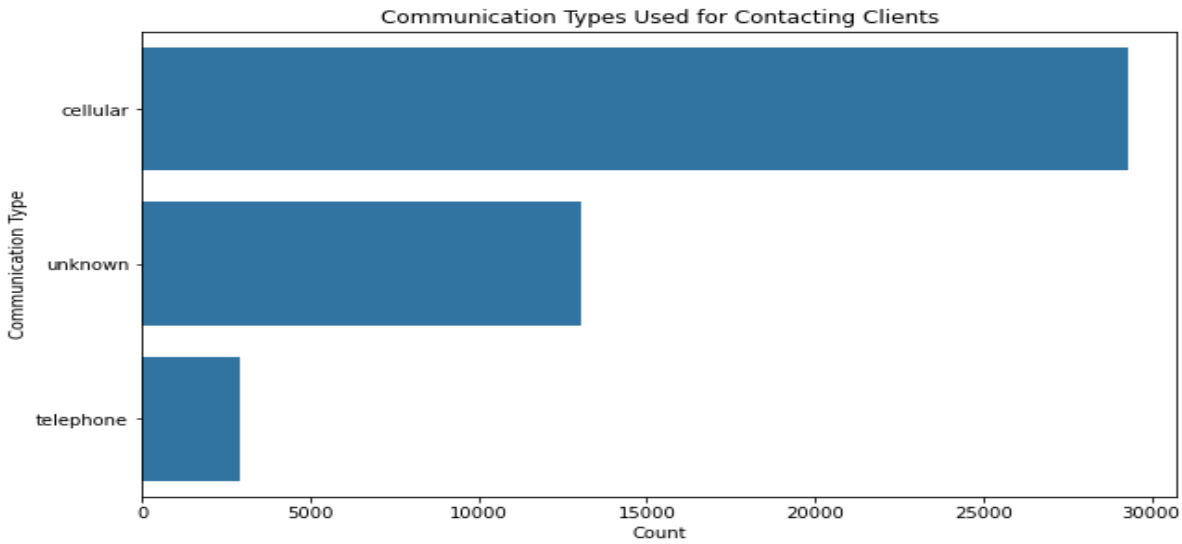
Communication type distribution

plt.figure(figsize=(10, 6)) sns.countplot(y='contact', data=df,

order=df['contact'].value_counts().index) plt.title('Communication Types

Used for Contacting Clients') plt.xlabel('Count') plt.ylabel('Communication

Type') plt.show()



Last contact day distribution

plt.figure(figsize=(10, 6)) sns.histplot(df['day'], bins=31,

kde=False) plt.title('Distribution of Last Contact Day of

the Month') plt.xlabel('Day of the Month')

plt.ylabel('Frequency') plt.show()

Last contact month distribution

plt.figure(figsize=(10, 6))

sns.countplot(y='month', data=df, order=df['month'].value_counts().index)

plt.title('Distribution of Last Contact Month') plt.xlabel('Count')

plt.ylabel('Month') plt.show()

Duration of last contact distribution

plt.figure(figsize=(10, 6)) sns.histplot(df['duration'],

bins=20, kde=True) plt.title('Distribution of

Duration of Last Contact') plt.xlabel('Duration

(seconds)') plt.ylabel('Frequency') plt.show()

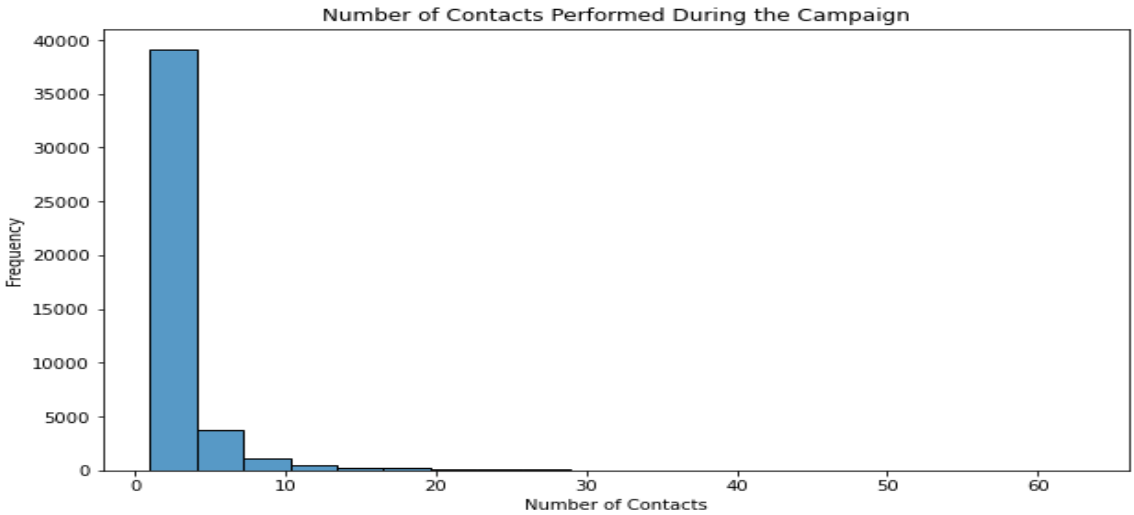
Number of contacts during the campaign

plt.figure(figsize=(10, 6)) sns.histplot(df['campaign'], bins=20,

kde=False) plt.title('Number of Contacts Performed During the

Campaign') plt.xlabel('Number of Contacts')

plt.ylabel('Frequency') plt.show()



Distribution of number of days passed since last contact from previous campaign

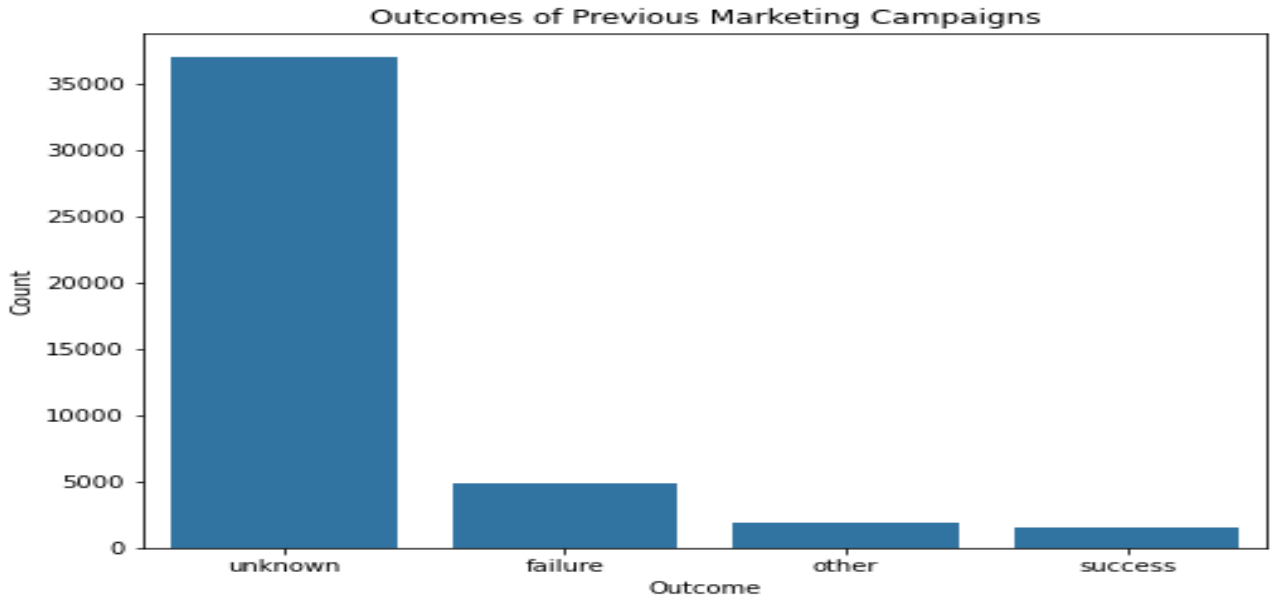
plt.figure(figsize=(10, 6)) sns.histplot(df['pdays'], bins=50, kde=False)

plt.title('Distribution of Number of Days Since Last Contact from Previous Campaign')

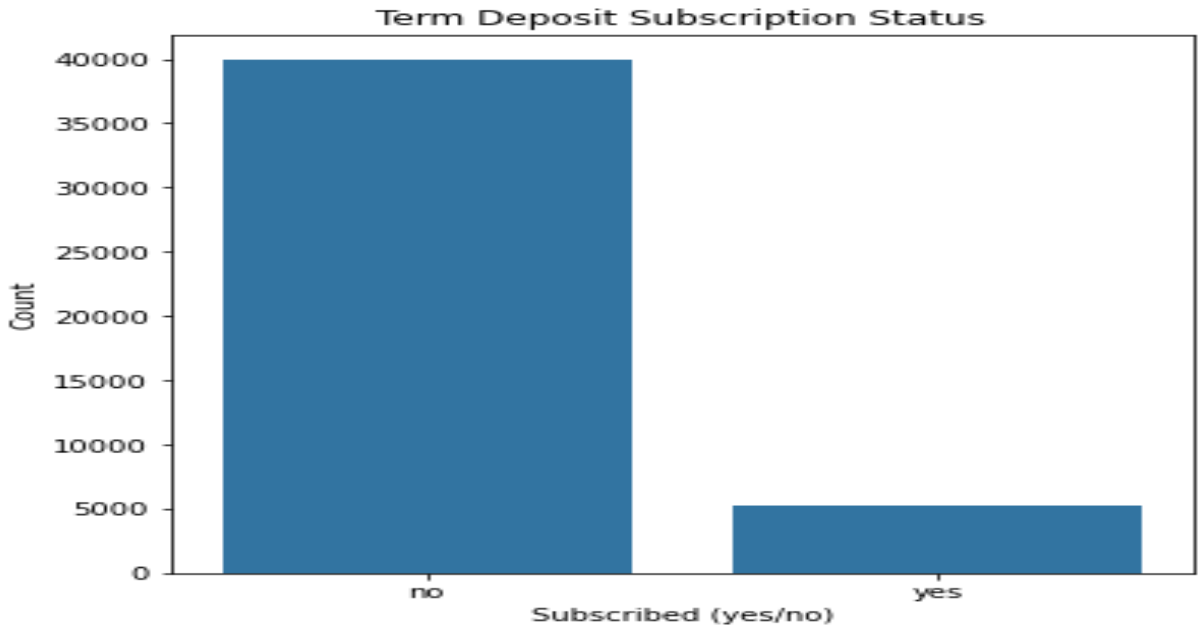
plt.xlabel('Number of Days') plt.ylabel('Frequency') plt.show()

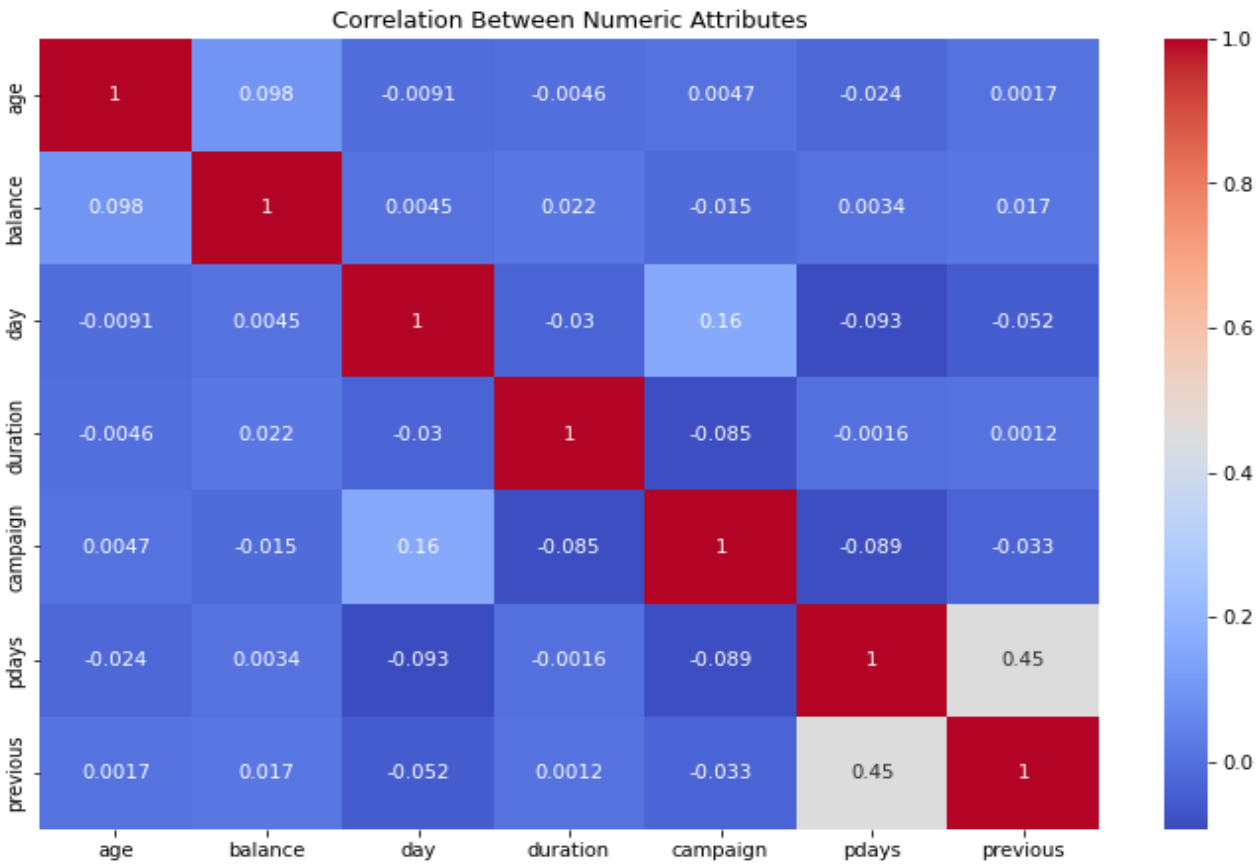
```
# Previous contacts distribution plt.figure(figsize=(10, 6))
sns.histplot(df['previous'], bins=20, kde=False) plt.title('Number of
Contacts Performed Before Current Campaign') plt.xlabel('Number
of Contacts') plt.ylabel('Frequency') plt.show()

# Previous outcome distribution
plt.figure(figsize=(8, 6))
sns.countplot(x='poutcome', data=df)
plt.title('Outcomes of Previous Marketing
Campaigns') plt.xlabel('Outcome')
plt.ylabel('Count') plt.show()
```




```
# Subscription to term deposit
plt.figure(figsize=(6, 6))
sns.countplot(x='y', data=df) plt.title('Term
Deposit Subscription Status')
plt.xlabel('Subscribed (yes/no)')
plt.ylabel('Count') plt.show()
```




```
# Select only numeric columns for correlation
numeric_df = df.select_dtypes(include='number')

# Plot the heatmap for correlation
plt.figure(figsize=(12, 8))
sns.heatmap(numeric_df.corr(), annot=True, cmap='coolwarm')
plt.title('Correlation Between Numeric Attributes') plt.show()
```